4.4 Trigonometric Form of a Complex Number

Name _____ Period ____ Date _____

Pre-Calculus 4.4 · Student sheet

I can…

- write a complex number in trigonometric form.
- write a complex number in trigonometric form in standard form.
- multiply and divide complex numbers in trigonometric form.

Vocabulary · fill in the missing word

Trigonometric form — $z =$ _____

Modulus — $r =$ _____ $= \sqrt{(a^2 + b^2)}$

Argument — the _____ θ, where $\tan\theta = \frac{b}{a}$

Part 1 · Distance and direction

Plot each number. Find r and θ, then check r cos θ and r sin θ.

Number	r	θ	r cos θ	r sin θ
$1 + i$				
$\sqrt{3} + i$				
-2				

Part 2 · Notes and examples (fill in each note, then work the example)

Trigonometric form: $a + bi = r(\cos\theta + i\sin\theta)$, with $r =$ _____ and $\tan\theta =$ _____.

Example 1

Write $-2 - 2\sqrt{3}\,i$ in trigonometric form.

Choosing θ: use the _____ of the point $a + bi$.

Example 2

Write $-1 + i$ in trigonometric form.

Example 3

Write $2\sqrt{2}\left(\cos\left(\frac{-\pi}{3}\right) + i\sin\left(\frac{-\pi}{3}\right)\right)$ in standard form.

Product: $z_1z_2 = r_1r_2[\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]$: multiply the moduli and _____ the arguments.

Example 4

Multiply: $2\left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3}\right) \cdot 8\left(\cos \frac{11\pi}{6} + i \sin \frac{11\pi}{6}\right)$

Quotient: divide the moduli and _____ the arguments.

Example 5

Divide: $24(\cos 300° + i \sin 300°) \div 8(\cos 75° + i \sin 75°)$

Part 3 · Partner practice

Partner 1

Write $1 - \sqrt{3}\ i$ in trigonometric form.

Partner 2

Multiply: $3(\cos 40° + i \sin 40°) \cdot 2(\cos 50° + i \sin 50°)$. Write the answer in standard form.